HPGC-Diff: Hybrid-Prior Guided Coupled Diffusion for Unsupervised Hyperspectral Image Super-Resolution

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***Abstract*—Fusing a low-resolution hyperspectral image (LR-HSI) with a high-resolution multispectral image (HR-MSI) is a widely adopted strategy for hyperspectral image super-resolution (HISR), for which diffusion models have recently shown strong potential. However, existing methods still suffer from two critical limitations: 1) spectral models based on multilayer perceptrons (MLPs) lack the necessary inductive bias for sequential data, structurally disregarding the local correlations between adjacent spectral bands and 2) spatial models struggle to simultaneously leverage the specific structures from the observed image with the generic priors learned from large-scale datasets. To overcome these challenges, we propose a hybrid-prior guided coupled diffusion (HPGC-Diff) for the unsupervised HISR task, which effectively integrates a low-rank prior, a spectral local correlation prior, and a generic prior. Leveraging the low-rank representation, we decompose the target HR-HSI into two low-dimensional components and establish separate processes for their joint reconstruction. Specifically, we design a 1-D U-Net spectral diffusion model that effectively learns the structured spectral distribution from the LR-HSI. For spatial modeling, we introduce a dual-source spatial model that integrates a generic prior from multiple pretrained diffusion models to provide a powerful generative capability, with conditional features extracted from the HR-MSI by a dedicated network to inject fine-grained structural details. Finally, spatial and spectral diffusion sampling is jointly guided and alternated with conditional feature optimization to ensure stable convergence under observational constraints. Extensive experiments on simulated and real-world datasets demonstrate that HPGC-Diff achieves superior performance compared to state-of-the-art methods.**

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***Index Terms*—Diffusion model, hyperspectral images (HSIs), image fusion, super-resolution.**

I. Introduction

HYPERSPECTRAL images (HSIs) capture the spectral reflectance of materials across hundreds of contiguous and narrow bands, making them highly valuable in applications, e.g., precision agriculture [1], [2], [3], mineral exploration [4], [5], and land cover classification [6], [7]. Recent advances in deep learning (DL) have further promoted HSIs in high-level interpretation tasks such as classification and object detection [8], [9], [10], [11], [12]. However, due to the inherent physical limitations of imaging systems, achieving high spatial and spectral resolution simultaneously remains a significant challenge. This tradeoff greatly limits the applicability of HSIs in tasks that require fine spatial detail [13]. To address this issue, a commonly adopted solution is to fuse a low-resolution HSI (LR-HSI) with a high-resolution multispectral image (HR-MSI) to generate a high-resolution HSI (HR-HSI) [14], [15], [16].

Over the past decade, numerous methods have been proposed for the hyperspectral and multispectral image (HSI-MSI) fusion task. These methods can be broadly categorized into two types: model-based methods and DL-based methods. Model-based methods typically formulate the fusion task as an inverse problem, regularizing the solution space with hand-crafted image priors [17]. Techniques such as sparse representation [18], [19], [20], low-rank representation [21], [22], [23], and tensor decomposition [24], [25], [26], [27] have been widely adopted to capture the intrinsic structure of the images. However, such hand-crafted priors are often insufficient to capture the complex and nonlinear feature distributions present in images. Furthermore, these methods often converge to suboptimal solutions during iterative optimization. In recent years, DL-based methods have attracted increasing attention for their powerful nonlinear modeling capabilities and end-to-end optimization frameworks. Consequently, researchers have proposed various fusion architectures based on convolutional neural networks (CNNs) [28], [29], [30], [31] and transformers [32], [33], [34], which aim to learn an end-to-end mapping from the observed images to the target image. Nevertheless, the majority of these methods are implemented within a supervised learning framework and are reliant on extensive paired

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training data. In practice, the prohibitive cost of acquiring paired data leads to data scarcity, which has become a critical bottleneck limiting their application.

More recently, diffusion models [35], [36], [37] have emerged as a new paradigm of deep generative models, demonstrating immense potential in the field of computer vision. In the domain of natural images, the capability of diffusion models to represent complex data distributions has driven exceptional performance in a diverse array of tasks, including image generation [38], [39], text-to-image generation [40], [41], and various inverse problems in image restoration [42], [43], [44]. Crucially, unlike generative adversarial networks (GANs), diffusion models offer a more stable training process. However, directly applying them to HSI-MSI fusion remains a challenging task. A key open question is how to jointly ensure spectral fidelity and the preservation of spatial details in the diffusion framework. Moreover, most existing diffusion-based approaches for HSI-MSI fusion [45], [46], [47], [48], [49] operate within a supervised learning framework, which severely constrains their practicality and generalizability. These limitations emphasize the urgent need for unsupervised HSI-MSI fusion as a direction of both theoretical importance and practical relevance. However, directly applying them to HSI-MSI fusion remains a challenging task. A key open question is how to jointly ensure spectral fidelity and the preservation of spatial details in the diffusion framework. Moreover, most existing diffusion-based approaches for HSI-MSI fusion [45], [46], [47], [48], [49] operate within a supervised learning framework, which severely constrains their practicality and generalizability.

Although several unsupervised diffusion model-based methods have been proposed in recent years, they still exhibit limitations in their spectral and spatial modeling. Existing methods [50], [51] employ diffusion models with multilayer perceptrons (MLPs) for spectral modeling. However, the fully connected structure of MLPs causes them to treat each spectral band as an independent feature, thereby structurally disregarding the sequential order and local correlations between bands. This inherent deficiency makes it difficult to effectively capture the absorption features governed by the physical properties of materials. In terms of spatial modeling, Zhu et al. [51] attempt to train a spatial diffusion model directly from the observed images, but the limited training samples often compromise the generative quality. On the other hand, Rui et al. [52] introduce a diffusion model pretrained on natural images to generate spatial structures. However, due to the lack of effective injection of structural information from the observed image, it struggles to recover fine-grained image details.

To address these limitations, we propose a hybrid-prior guided coupled diffusion (HPGC-Diff) for unsupervised HSI-MSI fusion, which effectively integrates a low-rank prior, a spectral local correlation prior, and a generic prior from large-scale datasets. Leveraging the low-rank representation, our approach jointly reconstructs an estimate of the HR-HSI by establishing separate processes for the decoupled spatial and spectral dimensions. Specifically, we move beyond the MLP-based designs by introducing a 1-D U-Net spectral diffusion model. Through its hierarchical convolutional design and self-attention mechanisms, this model can effectively capture the spectral continuity and long-range dependencies inherent in the observed LR-HSI. Our spatial modeling is driven by a dual-source architecture that integrates a generic prior from pretrained diffusion models to establish a robust generative foundation, alongside conditional features to inject deterministic local details, effectively bridging the distributional gap between the generic prior and the specific observation. To extract the conditional feature, we design a conditional feature extraction network (CFEN) that supplements spatial details and achieves maximum fidelity. Furthermore, we employ an alternating update strategy to ensure a robust and reliable reconstruction process. Finally, both branches are guided by a shared guidance function that enforces consistent convergence of the reverse diffusion trajectories through a coupled correction gradient. In summary, our contributions are as follows.

1) We propose an HPGC-Diff framework, which operates without reliance on external datasets. Leveraging the low-rank representation, it decomposes the target into two low-dimensional components and generates them via separate processes. A shared guidance function ensures cross-branch synergy for a coherent and collaborative reconstruction.
2) We design a 1-D U-Net spectral diffusion model for the spectral distribution. The proposed architecture is effective in capturing the spectral continuity and global relationships inherent in hyperspectral data, significantly outperforming conventional MLP-based designs.
3) We introduce a dual-source spatial model that integrates a generic prior from large-scale images with a conditional feature from the observation. A CFEN is designed to distill spatial structures and compensate for the absence of detail.
4) Extensive experiments on simulated and real-world datasets validate the effectiveness and superiority of our proposed method compared to state-of-the-art methods.

The remainder of this article is organized as follows. In Section II, we provide an introduction to the development of HSI-MSI fusion. Section III introduces the notations and preliminaries used throughout the article. In Section IV, the proposed HPGC-Diff is introduced in detail. Section V discusses the results of extensive evaluations. Finally, Section VI summarizes this article.

II. Relate Work

This section provides a structured overview of mainstream HSI-MSI fusion methods, categorizing them into model-based and DL-based approaches. Subsequently, special attention is devoted to the emerging class of DL approaches driven by diffusion models.

A. Model-Based Methods

Model-based methods represent the traditional paradigm for HSI-MSI fusion, formulating it as an ill-posed inverse problem regularized by hand-crafted priors that encode domain knowledge. A widely adopted strategy is to leverage the inherent sparsity of spectral signatures, modeling them as a sparse

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linear combination of atoms from a predefined dictionary. Kawakami et al. [18] focuses on learning a single spectral dictionary, which is then used to reconstruct the HR-HSI by estimating a sparse coefficient map constrained by the HR-MSI. In contrast, Han et al. [20] introduce an optimized twin-dictionary scheme, designed to simultaneously capture both generalized spectral signatures and high-frequency spatial details from residual information. Another class of priors leverages the inherent low-rank structure of HSIs, which assumes that the data lie within a low-dimensional subspace. A foundational approach, coupled nonnegative matrix factorization (CNMF) [53], jointly estimates low-rank endmembers and high-resolution abundances by factorizing the observations. To enhance robustness and flexibility, subsequent research has integrated this low-rank assumption into more sophisticated optimization frameworks. For instance, Simoes et al. [54] embed the subspace constraint within a convex formulation, regularizing the problem with vector total variation to preserve spatial edges. Zhang et al. [55] proposed a group spectral embedding regularizer, which complements the low-rank factorization by enforcing the constraint that spectrally similar bands share a common manifold structure. Xue et al. [23] learn to preserve both the nonlocal self-similarity across spatial patches and the global correlation among spectral bands within a unified variational model.

However, the matricization step required by these low-rank models can disrupt the HSI's intrinsic spatial–spectral structure. To overcome this limitation, several methods employ tensor decomposition. For instance, Li et al. [56] proposed a coupled sparse tensor factorization (CSTF) framework that represents the HSI via Tucker decomposition with a sparse core tensor. Dian et al. [57] incorporated a nonlocal self-similarity prior by assuming that similar image patches share common dictionaries. Building on a similar nonlocal principle, Xu et al. [58] extended this concept to a coupled model based on CP decomposition. Chen et al. [59] introduced a factor smoothed tensor ring (TR) decomposition, which regularizes the TR factors themselves to better preserve spatial–spectral continuity. In [27], a novel tensor double-factor decomposition is proposed, which represents the HSI as a contraction of a higher order spatial factor and a spectral factor to better preserve spatial structures.

B. DL-Based Methods

To overcome the limited expressiveness of hand-crafted priors, DL-based methods have emerged as the prevailing paradigm, learning an end-to-end mapping from the observed images to the target HR-HSI in a data-driven manner. These approaches can be broadly divided into two categories, i.e., supervised and unsupervised learning. Supervised methods learn powerful priors from large-scale paired datasets, leveraging their diversity to generalize effectively to unseen images. For example, a pioneering study by Yang et al. [60] introduced a two-branch CNN that jointly extracts features from the LR-HSI and the spatial neighborhood in the HR-MSI to perform fusion. In [61], a parallel spatial attention branch was designed to identify and enhance high-frequency details. Ran et al. [31] proposed a dedicated branch that decomposes the HR-MSI into a multiscale pyramid and then progressively guides the feature reconstruction. Echoing the widespread success of generative models in remote sensing [63], some works have incorporated GANs to further enhance the perceptual quality of the fused images. Xiao et al. [64] proposed a physics-based GAN where the generator was constrained by discriminators to ensure that its output conforms to the physical degradation model. Zhu et al. [65] introduced a lightweight adversarial network that leverages a novel quadtree implicit sampling strategy for high-fidelity spectral generation.

More recently, the transformer architecture has been introduced to this field, offering a distinct capability to model long-range dependencies. Initial approaches like Fusformer [32] leveraged the global self-attention of vision transformers to capture long-range interactions. Ma et al. [66] introduced a dual cross-attention mechanism for explicit feature exchange. To improve computational efficiency, a line of work has shifted to window-based self-attention, with methods like PSRT [33] using a shuffle operation and NGST-Net [67] employing an N-gram strategy to facilitate cross-window information flow. However, the effectiveness of these supervised paradigms critically relies on the distributional consistency between training and testing data. Even slight domain shifts can lead to pronounced performance degradation, thereby constraining their robustness and generalization in practice.

In light of the aforementioned challenges, unsupervised learning has emerged as an attractive solution for the fusion task. One prominent direction employs deep denoisers as implicit regularizers within iterative optimization. Dian et al. [28] implemented this by embedding a pretrained CNN denoiser into an ADMM solver, while Wang et al. [68] used an in situ self-supervised denoiser trained on the image's eigen-components. In parallel, another line of work integrates classical models into deep architectures: Zheng et al. [69] parameterized unknown degradation functions as learnable layers within a coupled autoencoder grounded in spectral unmixing theory, and Liu et al. [70] explicitly embedded a nonnegative matrix factorization model into an autoencoder's latent space. Beyond these, Li et al. [71] introduced a conceptual framework to synergistically fuse both external and internal learned priors, Wang et al. [72] proposed a variational optimization-guided two-stage network. Furthermore, Cao et al. [73] proposed a hybrid CNN-transformer architecture to capture global dependencies, and Li et al. [74] refined the deep image prior by substituting its random noise input with initializations derived from a zero-shot learning stage.

C. Diffusion Model-Based Methods

Building on their success in natural image generation and restoration, diffusion models have been explored as a promising paradigm for HSI-MSI fusion, with most existing approaches operating in a supervised framework. One major line of research centers on the denoising backbone and its conditioning mechanism. Shi et al. [45] established a foundational model by conditioning a U-Net backbone on concatenated HSI-MSI inputs. Wu et al. [46] introduced a conditional denoising transformer to better model global spatio-spectral dependencies. More recently, Cao et al. [49] refined the

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conditioning strategy through dedicated modulation modules, enabling disentanglement of coarse-grained style information and fine-grained wavelet details from the conditional signal. Another major direction explores innovations in the generative process framework itself. Dong et al. [48] developed an interpretable model that unfolds a coarse-to-fine optimization into the network, structuring the generation across hierarchical scales. Qu et al. [47] introduced a cycle-consistent framework to bilaterally learn the spatial and spectral super-resolution pathways, enforcing fidelity between the two domains.

To circumvent the requirement for paired ground-truth data, a nascent body of work has begun to adapt these models to an unsupervised framework. Liu et al. [50] embed a spectral diffusion model as a deep prior within a MAP estimation framework. Zhu et al. [51] train two separate diffusion models from the observed images. Rui et al. [52] leverage a pretrained diffusion model by projecting the HSI super-resolution (HISR) task into generating a low-rank base tensor. Nevertheless, untapped potential remains in the synthesis of priors, particularly in modeling the inductive biases of spectral data while synergizing powerful generic models with the observational context.

III. Notations and Preliminaries

A. Notations

Throughout this article, the symbols x, $\mathbf{x}$, $\mathbf{X}$, and $\mathcal{X}$ are used to represent scalars, vectors, matrices, and tensors, respectively. For the task of HSI-MSI fusion, the tensors $\mathcal{X} \in \mathbb{R}^{h\times w\times L}$, $\mathcal{Y} \in \mathbb{R}^{H\times W\times l}$, and $\mathcal{Z} \in \mathbb{R}^{H\times W\times L}$ denote the LR-HSI, HR-MSI, and target HR-HSI, respectively. The term $\mathbf{x}_{ij}$ refers to the spectral vector at spatial location (i, j) of the tensor $\mathcal{X}$. In addition, $\mathcal{Z} \times_3 \mathbf{R}$ denotes the mode-3 multiplication between the tensor $\mathcal{Z}$ and the matrix $\mathbf{R}$.

B. Diffusion Model

Diffusion models constitute a powerful class of probabilistic generative models that approximate complex data distributions $p(\mathbf{x}_0)$ by learning to reverse a progressive noising process. In the forward diffusion process, a noisy sample $\mathbf{x}_t$ at any timestep $t \in [1, T]$ can be analytically sampled from the clean input $\mathbf{x}_0$ using a predefined noise schedule $\{\beta_t\}_{t=1}^T$

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon} \tag{1}$$

where $\alpha_t = 1-\beta_t$, $\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$, and $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$. The reverse process aims to recover the clean sample $\mathbf{x}_0$ by starting from a variable $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$ and iteratively denoising it. Under the deterministic sampling framework of DDIM [36], the noise prediction is first utilized to approximate the clean data $\hat{\mathbf{x}}_0$, which subsequently determines the variable $\mathbf{x}_{t-1}$ via

$$\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \approx \boldsymbol{\epsilon} = \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\mathbf{x}_0}{\sqrt{1-\bar{\alpha}_t}} \tag{2}$$

$$\hat{\mathbf{x}}_0 = \frac{\mathbf{x}_t - \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)}{\sqrt{\bar{\alpha}_t}} \tag{3}$$

$$\mathbf{x}_{t-1} = \sqrt{\bar{\alpha}_{t-1}}\hat{\mathbf{x}}_0 + \sqrt{1-\bar{\alpha}_{t-1}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t). \tag{4}$$

In the continuous-time limit ($T \to \infty$), the discrete diffusion process converges to a process governed by a stochastic differential equation (SDE) [75]. This framework precisely models both the forward and reverse processes with a corresponding pair of SDEs

$$d\mathbf{x} = f(\mathbf{x}, t)\,dt + g(t)\,d\mathbf{w} \tag{5}$$

$$d\mathbf{x} = \left[f(\mathbf{x}, t) - g(t)^2 \nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)\right] dt + g(t)\,d\bar{\mathbf{w}} \tag{6}$$

where $f(\cdot, t)$ and $g(t)$ denote the drift and diffusion coefficients. The term $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$ is known as the score function. The standard Wiener processes for the forward and reverse directions are denoted by $\mathbf{w}$ and $\bar{\mathbf{w}}$, respectively.

C. Degradation Model

The fusion of LR-HSI $\mathcal{X}$ and HR-MSI $\mathcal{Y}$ is fundamentally a highly ill-posed inverse problem, with the objective of reconstructing latent HR-HSI from these observations. We assume that both observed images are generated from the same HR-HSI $\mathcal{Z}$ through distinct spatial and spectral degradation processes. Specifically, $\mathcal{X}$ is modeled as a spatially degraded version of $\mathcal{Z}$, a process comprising a downsampling operation followed by additive noise

$$\mathcal{X} = \tau(\mathcal{Z}) + \mathcal{N}_{\text{HSI}} \tag{7}$$

where $\tau(\cdot)$ represents the spatial degradation operator, with a scale factor s satisfying $H = s \cdot h$ and $W = s \cdot w$. The term $\mathcal{N}_{\text{HSI}}$ denotes the corresponding additive noise. On the other hand, $\mathcal{Y}$ is modeled as a spectrally degraded version of $\mathcal{Z}$. This process is represented by a linear transformation defined by the spectral response function (SRF), $\mathbf{R} \in \mathbb{R}^{l\times L}$, which projects the L hyperspectral bands onto the l multispectral bands ($l \ll L$)

$$\mathcal{Y} = \mathcal{Z} \times_3 \mathbf{R} + \mathcal{N}_{\text{MSI}} \tag{8}$$

where $\mathcal{N}_{\text{MSI}}$ is an additive noise term. Therefore, the central challenge lies in optimally estimating $\mathcal{Z}$ from the observed $\mathcal{X}$ and $\mathcal{Y}$ based on the known degradation model.

IV. Methodology

This section details the proposed HPGC-Diff framework, the overall architecture of which is illustrated in Fig. 1.

A. Guided Reverse Diffusion Process

While the central goal of HSI-MSI fusion is to estimate the latent HR-HSI, directly modeling such high-dimensional data is hampered by the curse of dimensionality and excessive computational costs. To circumvent this problem, we exploit the intrinsic low-rank structure of the HSI by decomposing it into two low-dimensional components: a spatial structure tensor $\mathcal{A} \in \mathbb{R}^{H\times W\times d}$ and a spectral basis $\mathbf{E} \in \mathbb{R}^{d\times L}$. In our approach, $\mathcal{A}$ is a composite formed by concatenating $\mathcal{M} \in \mathbb{R}^{H\times W\times l}$ extracted from the HR-MSI to preserve the structural details of the scene and $\mathcal{D} \in \mathbb{R}^{H\times W\times(d-l)}$ generated by diffusion models to provide a solid structural foundation. The HR-HSI is then reconstructed via a mode-3 multiplication

$$\mathcal{Z} = \mathcal{A} \times_3 \mathbf{E} = \text{Concat}(\mathcal{D}, \mathcal{M}) \times_3 \mathbf{E} \tag{9}$$

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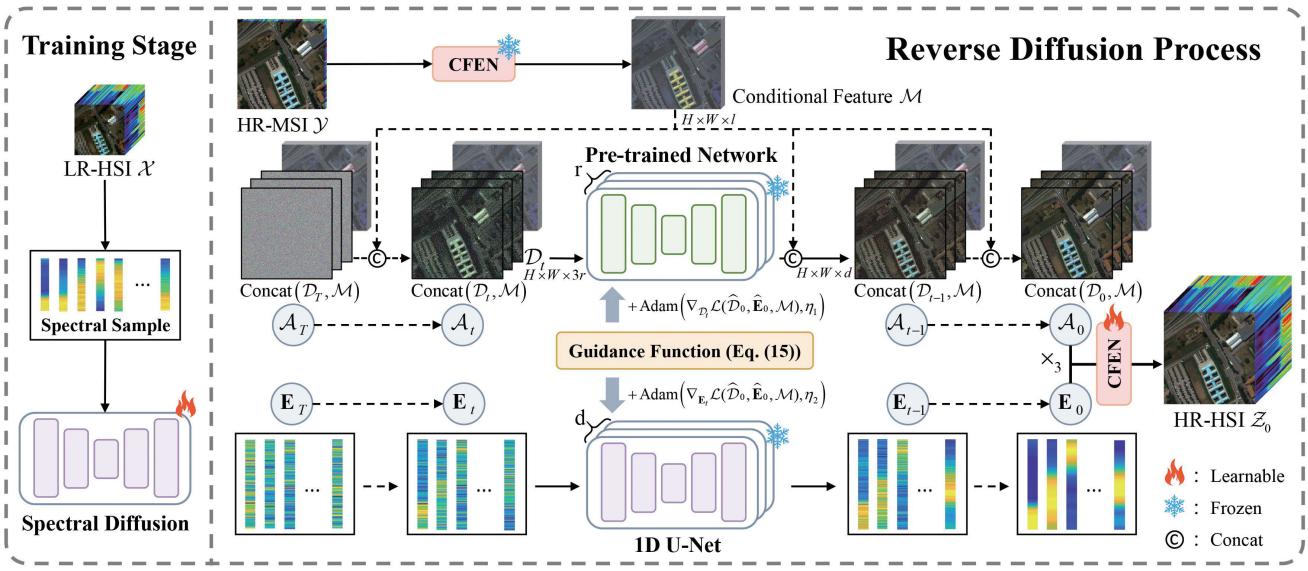


Fig. 1. Overall pipeline of the proposed HPGC-Diff, which comprises two stages. First, the 1-D U-Net spectral diffusion model is trained to capture intrinsic spectral distributions. Subsequently, the HR-HSI is reconstructed via an iterative refinement strategy. In each iteration, the diffusion model samples generative components under fixed conditional guidance, followed by a sequential update of the CFEN to refine the feature $\mathcal{M}$. The spatial structure tensor $\mathcal{A}$ formed by integrating $\mathcal{D}$ from pretrained diffusion models and $\mathcal{M}$ extracted by the CFEN.

where Concat$(\cdot,\cdot)$ denotes the concatenation operator applied along the feature dimension. Leveraging this composite decomposition, the original estimation problem is reformulated as a joint inference task. To facilitate stable optimization, we adopt an iterative alternating update strategy. During the diffusion sampling phase, the conditional feature $\mathcal{M}$ is treated as a fixed input rather than being jointly estimated. This design decouples conditional feature learning from diffusion sampling and reduces the generative task to a joint inference over the diffusion-based components $\mathcal{D}$ and $\mathbf{E}$. Under Bayesian inference, this joint estimation problem is equivalent to sampling from the joint posterior distribution $p(\mathcal{D}_t, \mathbf{E}_t|\mathcal{X}, \mathcal{Y}, \mathcal{M})$. Given that $\mathcal{D}_t$ and $\mathbf{E}_t$ are modeled by separate networks, it is reasonable to assume their priors are mutually independent. According to Bayes' rule, this posterior can be decomposed into the product of a likelihood term and two prior terms

$$p\left(\mathcal{D}_t, \mathbf{E}_t|\mathcal{X}, \mathcal{Y}, \mathcal{M}\right) \propto p\left(\mathcal{X}, \mathcal{Y}, \mathcal{M}|\mathcal{D}_t, \mathbf{E}_t\right) p\left(\mathcal{D}_t\right) p\left(\mathbf{E}_t\right). \quad (10)$$

Following the general reverse SDE in (6), we can derive the specific SDEs for $\mathcal{D}$ and $\mathbf{E}$. According to Bayes' rule in (10), the score functions $\nabla_{\mathcal{D}_t} \log p_t(\mathcal{D}_t, \mathbf{E}_t|\mathcal{X}, \mathcal{Y}, \mathcal{M})$ and $\nabla_{\mathbf{E}_t} \log p_t(\mathcal{D}_t, \mathbf{E}_t|\mathcal{X}, \mathcal{Y}, \mathcal{M})$ can be decomposed into the sum of a joint posterior score and a prior score. However, directly estimating the joint posterior score from $\mathcal{D}_t$ and $\mathbf{E}_t$ is intractable. Therefore, following the principles of diffusion posterior sampling [76], the overall score function is asymptotically equivalent to

$$\begin{aligned}
&\nabla_{\mathcal{D}_t} \log p_t\left(\mathcal{D}_t, \mathbf{E}_t|\mathcal{X}, \mathcal{Y}, \mathcal{M}\right) \\
&= \nabla_{\mathcal{D}_t} \log p_t\left(\mathcal{X}, \mathcal{Y}, \mathcal{M}|\mathcal{D}_t, \mathbf{E}_t\right) + \nabla_{\mathcal{D}_t} \log p_t\left(\mathcal{D}_t\right) \\
&\simeq \nabla_{\mathcal{D}_t} \log p\left(\mathcal{X}, \mathcal{Y}, \mathcal{M}|\hat{\mathcal{D}}_0, \hat{\mathbf{E}}_0\right) + \nabla_{\mathcal{D}_t} \log p_t\left(\mathcal{D}_t\right) \quad (11) \\
&\nabla_{\mathbf{E}_t} \log p_t\left(\mathcal{D}_t, \mathbf{E}_t|\mathcal{X}, \mathcal{Y}, \mathcal{M}\right) \\
&= \nabla_{\mathbf{E}_t} \log p_t\left(\mathcal{X}, \mathcal{Y}, \mathcal{M}|\mathcal{D}_t, \mathbf{E}_t\right) + \nabla_{\mathbf{E}_t} \log p_t\left(\mathbf{E}_t\right) \\
&\simeq \nabla_{\mathbf{E}_t} \log p\left(\mathcal{X}, \mathcal{Y}, \mathcal{M}|\hat{\mathcal{D}}_0, \hat{\mathbf{E}}_0\right) + \nabla_{\mathbf{E}_t} \log p_t\left(\mathbf{E}_t\right) \quad (12)
\end{aligned}$$

where $\hat{\mathcal{D}}_0$ and $\hat{\mathbf{E}}_0$ are the estimates of the final components $\mathcal{D}_0$ and $\mathbf{E}_0$ at timestep t. Following the reconstruction formulation in (3), these clean components are explicitly estimated via

$$\hat{\mathcal{D}}_0 = \frac{\mathcal{D}_t - \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_{c_\phi}\left(\mathcal{D}_t, t\right)}{\sqrt{\bar{\alpha}_t}} \quad (13)$$

$$\hat{\mathbf{E}}_0 = \frac{\mathbf{E}_t - \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}_{s_\theta}\left(\mathbf{E}_t, t\right)}{\sqrt{\bar{\alpha}_t}} \quad (14)$$

where $\boldsymbol{\epsilon}_{c_\phi}(\mathcal{D}_t, t)$ and $\boldsymbol{\epsilon}_{s_\theta}(\mathbf{E}_t, t)$ denote the noise predicted by the spatial network c_ϕ and the spectral network s_θ, respectively. These estimates allow us to reframe the intractable likelihood gradient problem on $\mathcal{D}_t$ and $\mathbf{E}_t$ as a tractable one defined on $\hat{\mathcal{D}}_0$ and $\hat{\mathbf{E}}_0$. The guidance function $\mathcal{L}(\hat{\mathcal{D}}_0, \hat{\mathbf{E}}_0, \mathcal{M})$ is then given by

$$\begin{aligned}
\mathcal{L}\left(\hat{\mathcal{D}}_0, \hat{\mathbf{E}}_0, \mathcal{M}\right) &= \log p\left(\mathcal{X}, \mathcal{Y}, \mathcal{M}|\hat{\mathcal{D}}_0, \hat{\mathbf{E}}_0\right) \\
&:= -\gamma_1 \left\|\tau\left(\text{Concat}\left(\hat{\mathcal{D}}_0, \mathcal{M}\right) \times_3 \hat{\mathbf{E}}_0\right) - \mathcal{X}\right\|_F^2 \\
&\quad - \gamma_2 \left\|\text{Concat}\left(\hat{\mathcal{D}}_0, \mathcal{M}\right) \times_3 \hat{\mathbf{E}}_0 \times_3 \mathbf{R} - \mathcal{Y}\right\|_F^2
\end{aligned} \quad (15)$$

where γ_1 and γ_2 are the weighting parameters. The ability to estimate the prior scores for $\nabla_{\mathcal{D}_t} \log p_t(\mathcal{D}_t)$ and $\nabla_{\mathbf{E}_t} \log p_t(\mathbf{E}_t)$ with their respective networks allows us to tackle the generative part of the fusion task with two parallel and tractable diffusion processes. Building on this framework, we introduce a gradient-guided reverse sampling scheme to synergistically generate $\mathcal{D}_t$ and $\mathbf{E}_t$. Starting from independent Gaussian noise $\mathcal{D}_T$ and $\mathbf{E}_T$, this process iteratively refines these components over T steps by discretizing their respective reverse SDEs following (4). At each step, the spatial noise prediction network $c_\phi(\mathcal{D}_t, t)$ and the spectral noise prediction network $s_\theta(\mathbf{E}_t, t)$ first estimate the inherent noise components. Then, we compute the intermediate unguided states $\mathcal{D}_{t-1}^*$ and $\mathbf{E}_{t-1}^*$ using (4) based on the predicted clean components $\hat{\mathcal{D}}_0$ and $\hat{\mathbf{E}}_0$

$$\mathcal{D}_{t-1}^* = \sqrt{\bar{\alpha}_{t-1}}\hat{\mathcal{D}}_0 + \sqrt{1-\bar{\alpha}_{t-1}}\boldsymbol{\epsilon}_{c_\phi}\left(\mathcal{D}_t, t\right) \quad (16)$$

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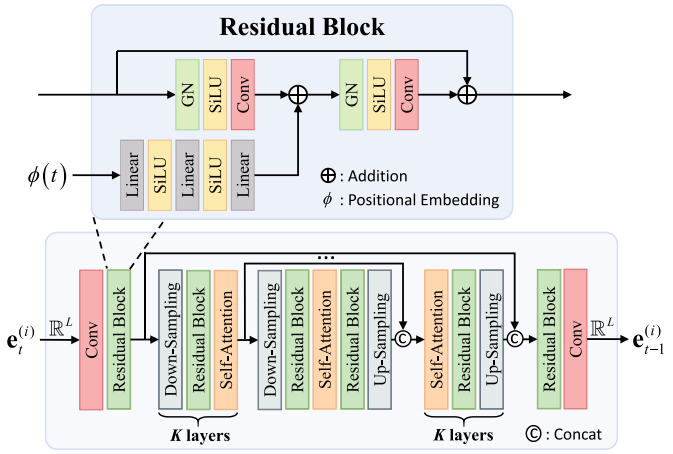


Fig. 2. Proposed 1-D U-Net architecture of the spectral diffusion model.

$$\mathbf{E}_{t-1}^{*} = \sqrt{\bar{\alpha}_{t-1}}\hat{\mathbf{E}}_0 + \sqrt{1-\bar{\alpha}_{t-1}}\boldsymbol{\epsilon}_{s_\theta}(\mathbf{E}_t, t). \tag{17}$$

Subsequently, the guidance function in (15) is applied to refine these intermediate states. Since this refinement corresponds to gradient-based optimization of the diffusion variables under observation guidance, we employ the Adam optimizer to enhance optimization stability [51], [77], [78]

$$\mathcal{D}_{t-1} = \mathcal{D}_{t-1}^{*} + \text{Adam}\left(\nabla_{\mathcal{D}_t}\mathcal{L}\left(\hat{\mathcal{D}}_0, \hat{\mathbf{E}}_0, \mathcal{M}\right), \eta_1\right) \tag{18}$$

$$\mathbf{E}_{t-1} = \mathbf{E}_{t-1}^{*} + \text{Adam}\left(\nabla_{\mathbf{E}_t}\mathcal{L}\left(\hat{\mathcal{D}}_0, \hat{\mathbf{E}}_0, \mathcal{M}\right), \eta_2\right) \tag{19}$$

where $\text{Adam}(\cdot, \eta)$ denotes the update step computed by the Adam optimizer with guidance step size η.

B. 1-D U-Net Spectral Diffusion Model

Accurately modeling the spectral basis distribution $p(\mathbf{E}_0)$ is a cornerstone of our framework. Notably, spectral signatures are not arbitrary vectors in the ambient space $\mathbb{R}^L$, but strong structural regularities as 1-D signals. This intrinsic structure arises as the physicochemical properties of materials yield inherently smooth spectral signatures with strong local correlations, which are subsequently preserved and reinforced by the physics of the imaging process. Consequently, we posit that spectral signals are highly concentrated on a low-dimensional data manifold with an intrinsic dimensionality much lower than L.

Algorithm 1 Training Procedure of Spectral Diffusion

Input: LR-HSI $\mathcal{X}$

1: Initialize network parameters θ;
2: **repeat**
3: Sample a spectral vector $\mathbf{e}_0 \sim \mathcal{X}$;
4: $t \sim \text{Uniform}(\{1, \ldots, T\})$;
5: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$;
6: Take gradient descent step on
7: $\nabla_\theta \|\boldsymbol{\epsilon} - s_\theta(\sqrt{\bar{\alpha}_t}\mathbf{e}_0 + \sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon}, t)\|^2$;
8: **until** converged

C. Training Paradigm

To learn this low-dimensional spectral manifold, we adopt a self-supervised training paradigm built on the assumption that the spatially blurred LR-HSI preserves high-fidelity spectral signatures. This allows the spectral network $s_\theta(\mathbf{E}_t, t)$ to be trained entirely without external data, which distinguishes it from previous methods like SDP [50]. Specifically, we construct an internal training dataset of $h \times w$ samples by treating the spectral vector $\mathbf{x}_{ij} \in \mathbb{R}^L$ of each pixel in the observed LR-HSI as an independent sample. Subsequently, the spectral network is optimized by minimizing the mean squared error (mse) between the actual added noise and the noise predicted by the network, following the training procedure detailed in Algorithm 1.

D. Model Architecture

The noise prediction network for our spectral diffusion model is implemented as a 1-D U-Net, explicitly incorporating the sequential inductive biases of spectral data, as shown in Fig. 2. In contrast to conventional MLP-based designs that flatten each spectrum into independent features, our architecture is tailored to capture both the local continuity and long-range dependencies intrinsic to hyperspectral signatures. The network backbone employs a hierarchical encoder–decoder architecture characterized by K downsampling and upsampling stages. Each stage comprises a time-aware residual block, where the diffusion timestep t is injected through linear embeddings added to intermediate feature representations. While these residual blocks effectively capture local spectral correlations through 1-D convolutions, their receptive fields are inherently constrained. To mitigate this limitation, self-attention modules [79] are embedded within the deeper layers to capture long-range spectral dependencies across distant bands. By synergizing local convolutional modeling with global attention mechanisms, the network learns a hierarchical representation that fuses fine-grained spectral details with global contextual information. Furthermore, to accommodate spectral lengths that are indivisible by the total downsampling factor, an adaptive padding strategy is implemented during decoding. Specifically, zero-padding is applied to the spectral tail of upsampled features to guarantee precise alignment with the corresponding encoder features prior to skip-connection concatenation.

Consequently, we estimate the spectral basis $\mathbf{E}$ using d independent spectral diffusion models. Correspondingly, the uncorrected spectral noise prediction network is defined as

$$s_\theta(\mathbf{E}_t, t) = \text{Concat}\left(s_\theta\left(\mathbf{e}_t^{(1)}, t\right), \ldots, s_\theta\left(\mathbf{e}_t^{(d)}, t\right)\right) \tag{20}$$

where $\{\mathbf{e}_t^{(i)}\}_{i=1}^{d}$ denotes the set of d independent spectral basis vectors.

E. Dual-Source Spatial Model

The necessity of our dual-source spatial modeling arises from the inherent limitations of relying on a single source of spatial information for high-fidelity reconstruction. Although diffusion priors capture complex data distributions, relying on them in isolation risks distributional discrepancy and structural deviations from the target scene. In contrast, spatial cues extracted directly from the HR-MSI provide strong

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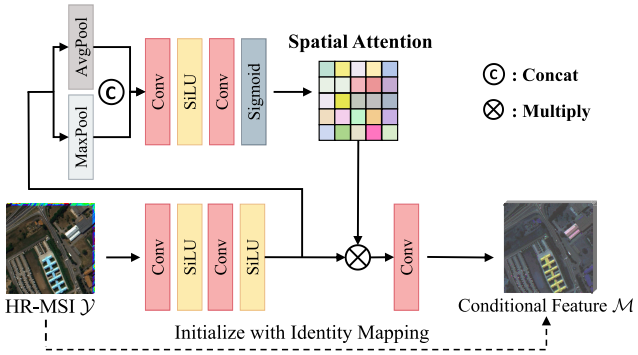


Fig. 3. CFEN.

local consistency with the observations but lack sufficient representational capacity to support the reconstruction of high-dimensional hyperspectral structures. To resolve this, we propose a dual-source spatial model that synergizes generic priors with conditional features to bridge this gap.

First, we leverage a diffusion model pretrained on large-scale remote sensing images to serve as a generic structural prior. However, the complex spatial heterogeneity inherent in hyperspectral scenes means that a single generative model [52], [80], initiated from one noise vector, risks converging to a restricted representation that fails to capture the full diversity of plausible material layouts and textures. To mitigate this limitation and construct a richer prior better suited to HSI's complexity, we employ an ensemble of r independent diffusion models that operate in parallel. During the reverse process, each model is initialized with a unique random noise sample to generate a distinct three-channel spatial feature map. The uncorrected spatial noise prediction network $c_\phi(\mathcal{D}_t, t)$ is formulated as follows:

$$c_\phi(\mathcal{D}_t, t) = \text{Concat}\left(c_\phi\left(\mathcal{D}_t^{(1)}, t\right), \ldots, c_\phi\left(\mathcal{D}_t^{(r)}, t\right)\right) \tag{21}$$

where $\mathcal{D}_t^{(i)} \in \mathbb{R}^{H\times W\times 3}$ denotes the noisy input at timestep t for the ith model.

While the pretrained ensemble offers a powerful generic prior for structural plausibility, its inherent agnosticism to the specific spatial layout of the target scene creates a distributional gap. This issue is further exacerbated in the unsupervised setting, where the lack of training data precludes effective fine-tuning. To improve spatial fidelity and alignment with the observation, we incorporate a conditional feature map extracted from the HR-MSI into the reverse diffusion process as a fixed condition. This conditional pathway compensates for the distributional discrepancy between the pretrained diffusion prior and the target scene by establishing a synergistic division of labor. The conditional pathway directly provides the specific details that are often absent in the generic prior, thereby alleviating the generative burden on the diffusion model and enabling it to focus on synthesizing a globally coherent and structurally plausible backbone.

Specifically, we introduce a dedicated CFEN to transform the HR-MSI into a more expressive feature space. The CFEN, implemented as a lightweight convolutional network with spatial attention, learns to extract high-level features optimized for complementing the reverse diffusion process with high-fidelity spatial information. The spatial attention mechanism plays a critical role in this design by adaptively focusing on informative regions that contribute most to structural restoration. In the proposed iterative optimization framework, the output of the CFEN serves as a conditional input to the diffusion model during each reverse sampling stage. At the first outer-loop iteration, the CFEN is initialized as an identity mapping to prevent the introduction of randomly transformed conditional features. After the reverse diffusion process produces the generative components $\mathcal{D}_0$ and $\mathbf{E}_0$, the CFEN parameters are subsequently updated by minimizing the reconstruction loss $\mathcal{L}(\mathcal{D}_0, \mathbf{E}_0, \mathcal{M})$. The detailed architecture of the CFEN is illustrated in Fig. 3.

Algorithm 2 Proposed Method (HPGC-Diff)

Input: LR-HSI $\mathcal{X}$. HR-MSI $\mathcal{Y}$. Total diffusion steps T.Weighting parameters γ_1, γ_2. Guidance step sizes η_1, η_2.Pre-trained spatial network c_ϕ. The number of generators r.

Output: HR-HSI $\mathcal{Z}_0$

```
Train the spectral network s_θ using X;
Initialize M as Y;
repeat
    for t = T, T − 1, ..., 1 do
        Estimate D̂_0, Ê_0 by Eq. (13), Eq. (14);
        Calculate ∇L(D̂_0, Ê_0, M) by Eq. (15);
        Calculate D*_{t−1}, E*_{t−1} by Eq. (16), Eq. (17);
        Update D_{t−1}, E_{t−1} by Eq. (18), Eq. (19);
    end for
    Update M by minimizing Eq. (15);
until converged
return Z_0 = A_0 ×_3 E_0 = Concat(D_0, M) ×_3 E_0.
```

As a result, the spatial tensor $\mathcal{A}$ comprises $d = 3r + l$ features, integrating the diffusion-generated components with the conditional feature

$$\mathcal{A}_t = \text{Concat}\left(\mathcal{D}_t^{(1)}, \mathcal{D}_t^{(2)}, \ldots, \mathcal{D}_t^{(r)}, \mathcal{M}\right). \tag{22}$$

The proposed HPGC-Diff is summarized in Algorithm 2.

V. Experiment

In this section, we comprehensively evaluate the proposed HPGC-Diff by conducting comparative experiments against various methods on three simulated datasets and one real-world dataset. Both quantitative and qualitative results are reported to demonstrate the effectiveness of our method. Then, we present the results of the ablation study and analyze the sensitivity of hyperparameters.

A. Datasets and Implementation Details

To evaluate the performance of the proposed method, we conduct experiments on both simulated and real-world data. Specifically, the Pavia University, Chikusei, and WDC datasets are used for simulation experiments, while the 2018 IEEE GRSS Data Fusion Contest (DFC2018) Houston dataset is employed for a real-world data experiment. Pavia University

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is a classic urban dataset acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over Pavia, northern Italy. The image has dimensions of $610 \times 340 \times 103$ with a spatial resolution of 1.3 m. It originally comprises 115 spectral bands spanning the 430–860-nm wavelength range, from which 12 noisy and water absorption bands are removed for analysis. Chikusei is an airborne dataset captured by the Headwall Hyperspec-VNIR-C sensor over an agricultural and urban area in Chikusei, Japan. The full image has dimensions of $2517 \times 2335 \times 110$ with a spatial resolution of 2.5 m. It originally consists of 128 spectral bands covering the 363–1018-nm wavelength range, from which 18 noisy bands are discarded prior to analysis. WDC was acquired by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) airborne sensor over the Washington DC Mall area. This dataset has spatial dimensions of 1280×307 pixels and a spectral range from 400 to 2400 nm. After removing 19 bands affected by atmospheric and water absorption, 191 spectral bands are retained for evaluation. The DFC2018 Houston dataset was acquired by the ITRES CASI-1500 sensor over the University of Houston campus and the surrounding urban area. It features a high spatial resolution of 1.0 m. The hyperspectral data has dimensions of $1202 \times 4172 \times 48$, covering the 380–1050-nm wavelength range.

For the simulation experiments, a clean image patch of size 256×256 is cropped from each of the three simulation datasets as the ground-truth. To generate the degraded inputs for our model, both spatial and spectral degradations are simulated on these clean images. Specifically, all LR-HSIs are generated using a uniform spatial downsampling factor of 4. The corresponding HR-MSIs are produced by spectrally degrading the ground-truth using known SRFs. For the Pavia University, Chikusei, and WDC datasets, the number of spectral bands in the resulting HR-MSIs is set to 4, 5, and 6, respectively. To emulate realistic sensing conditions, all simulated observations are corrupted by additive Gaussian noise with a signal-to-noise ratio (SNR) of 35 dB. For the real-world experiment on the Houston dataset, a 64×64 patch from the original HSI is used as the LR-HSI, while the corresponding RGB image is processed to 256×256 to serve as the HR-MSI.

All experiments are conducted on the PyTorch and MATLAB (R2022a) platform, with an Intel Core i7-14700K CPU, an NVIDIA GeForce RTX 4090 GPU, and 64-GB RAM. For the proposed 1-D U-Net spectral diffusion model, we set the number of hierarchical stages to $K = 2$. Regarding its self-supervised training, we utilized the AdamW optimizer with a learning rate of $1e^{-4}$ and a batch size of 128. The training process was conducted for 50 000 epochs to ensure convergence. For the generic spatial prior, we employ the pretrained diffusion model specifically built for remote sensing images [81]. For all experiments, we set the number of total diffusion steps to $T = 400$. The weighting parameters are $\gamma_1 = \gamma_2 = 1$, and the guidance step sizes are $\eta_1 = 0.05$ and $\eta_2 = 0.03$. The number of pretrained generators for the generic prior r is set to 3. The CFEN is optimized for 8000 epoches using the Adam optimizer with a learning rate of $5e^{-4}$. The overall framework executes 3 outer-loop iterations for convergence.

TABLE I
QUANTITATIVE COMPARISON ON THE PAVIA UNIVERSITY. THE BEST RESULTS ARE IN BOLD AND THE SECOND BEST ARE UNDERLINED

Method	PSNR ↑	SAM ↓	ERGAS ↓	SSIM ↑	RMSE ↓
CNMF	37.78	3.67	2.34	0.965	0.0142
HySure	38.03	3.89	2.39	0.963	0.0139
CSTF	39.30	3.46	2.02	0.968	0.0122
HyCoNet	40.17	2.83	1.76	0.968	0.0110
MIAE	41.29	2.87	1.73	0.972	0.0102
EDIP-Net	41.22	<u>2.53</u>	1.61	0.972	0.0096
PLRDiff	40.36	3.34	1.90	0.971	0.0113
ARGS-Diff	<u>42.05</u>	2.77	<u>1.53</u>	<u>0.975</u>	<u>0.0093</u>
HPGC-Diff	**43.96**	**2.23**	**1.34**	**0.982**	**0.0078**

B. Competing Methods and Evaluation Metrics

We compare the proposed method with eight state-of-the-art approaches. The traditional methods include CNMF [53], HySure [54], and CSTF [56]. The DL-based methods include HyCoNet [69], MIAE [70], and EDIP-Net [74]. The diffusion model-based methods include PLRDiff [52] and ARGS-Diff [51]. To ensure a fair comparison between blind and nonblind methods, all approaches are evaluated under a unified degradation setting. Specifically, the spatial and spectral degradation operators are estimated using the semi-blind degradation prediction network from ARGS-Diff [51] and are then fixed and shared across all methods. For quantitative evaluation, we adopt five widely used metrics: peak SNR (PSNR), spectral angle mapper (SAM), erreur relative globale adimensionnelle de synthèse (ERGAS), structural similarity index measure (SSIM), and root mean squared error (RMSE).

C. Experimental Results

1) Pavia University: The visual results and the corresponding error maps for each method are shown in Fig. 4. Traditional methods like CNMF and HySure suffer from spatial blurring and distortion, failing to reconstruct sharp edges and fine textures. Although CSTF improves structural correlation, its performance is ultimately constrained by its reliance on handcrafted priors. Data-driven approaches, encompassing both DL and diffusion-based models, achieve substantially higher fidelity. However, they still exhibit subtle structural inconsistencies and textural imperfections, particularly regarding the vehicles in the lower left region. In contrast, our proposed HPGC-Diff faithfully reconstructs these fine-grained details, a visual superiority that is corroborated by the quantitative dominance reported in Table I. Notably, it achieves a PSNR of 43.96 dB, marking a substantial margin of 1.91 dB over the second-best method, ARGS-Diff. This leading performance extends across all other key metrics. This consistent numerical superiority, especially over recent advanced methods like EDIP-Net and ARGS-Diff, underscores the efficacy of our proposed approach in achieving a more accurate and robust reconstruction with superior spatial and spectral fidelity. The spectral profile of PSNR in Fig. 5 further confirms this superiority, revealing a performance lead for our method across all bands.

2) Chikusei: Fig. 6 illustrates the visual comparison and corresponding error maps for all competing methods. A

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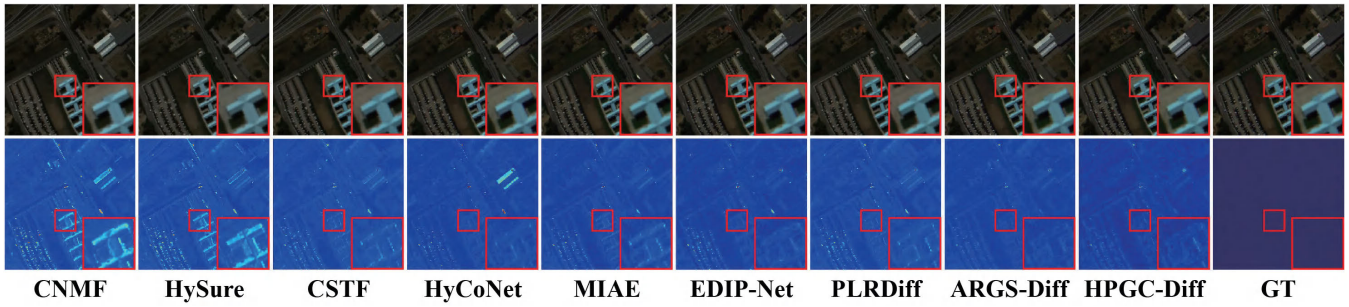


Fig. 4. Qualitative comparison of different fusion methods on the Pavia University dataset. The top row displays the visual fusion results (R:61, G:36, and B:10). The bottom row shows the corresponding reconstruction error maps, where darker colors indicate smaller errors.

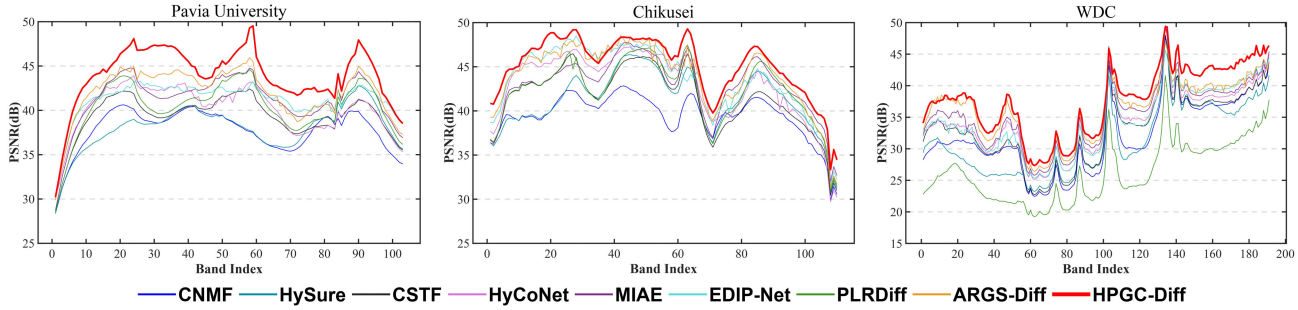


Fig. 5. PSNR value of each band in different HSIs.

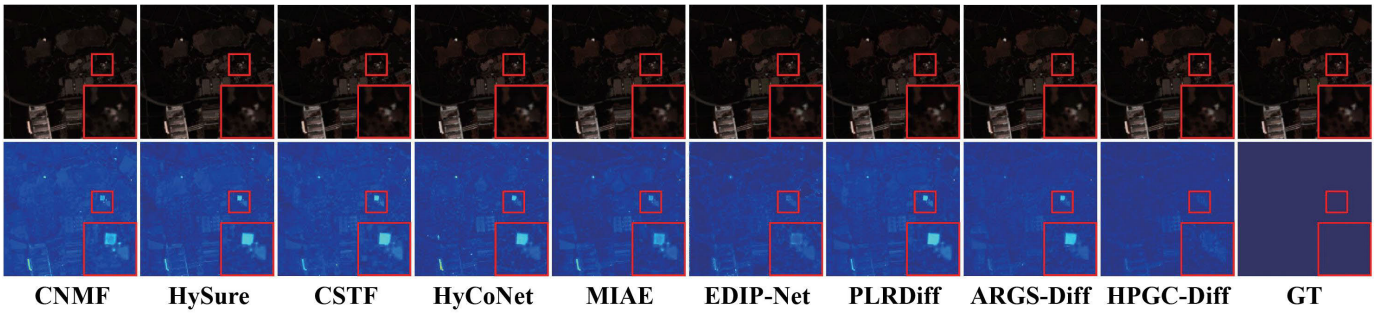


Fig. 6. Qualitative comparison of different fusion methods on the Chikusei dataset. The top row displays the visual fusion results (R:56, G:23, and B:13). The bottom row shows the corresponding reconstruction error maps, where darker colors indicate smaller errors.

qualitative analysis clearly demonstrates the superiority of our proposed HPGC-Diff. Although competing methods deliver reasonably high-quality reconstructions, closer inspection reveals clear disparities in handling complex structural details. For instance, in the magnified regions showing the building's roof, the error maps of competing approaches exhibit prominent high-intensity signals. This indicates a significant loss of structural fidelity and the introduction of geometric distortions. In stark contrast, the corresponding area in the error map of our HPGC-Diff is almost entirely dark, reflecting its superior ability to accurately capture fine-grained structural details and maintain high reconstruction fidelity. This qualitative advantage is consistently reflected in the quantitative evaluation, as shown in Table II, where HPGC-Diff outperforms all methods across every metric. Fig. 5 further highlights the fidelity of our spatial reconstruction.

3) WDC: For qualitative evaluation, the reconstructed images and corresponding error maps of all methods are depicted in Fig. 7. Traditional methods struggle with the intricate urban layout, resulting in severe spatial degradation and the loss of structural integrity. Furthermore, the high spectral dimensionality of the WDC dataset, which comprises 191 spectral bands, poses a significant challenge even for some advanced data-driven approaches. This limitation is clearly observed in PLRDiff, whose reconstruction exhibits not only significant visual distortions but also a glaringly bright error map. Conversely, our proposed HPGC-Diff demonstrates

TABLE II

QUANTITATIVE COMPARISON ON THE CHIKUSEI. THE BEST RESULTS ARE IN BOLD AND THE SECOND BEST ARE UNDERLINED

Method	PSNR ↑	SAM ↓	ERGAS ↓	SSIM ↑	RMSE ↓
CNMF	38.39	2.43	3.07	0.976	0.0118
HySure	40.06	2.19	2.92	0.979	0.0103
CSTF	40.39	2.09	2.41	0.977	0.0102
HyCoNet	41.75	1.88	2.06	0.981	0.0095
MIAE	41.84	1.75	2.33	0.982	0.0102
EDIP-Net	42.59	<u>1.66</u>	1.85	<u>0.983</u>	0.0082
PLRDiff	41.25	2.02	2.37	0.979	0.0093
ARGS-Diff	<u>42.96</u>	1.79	<u>1.80</u>	0.979	<u>0.0081</u>
HPGC-Diff	**44.05**	**1.50**	**1.57**	**0.984**	**0.0069**

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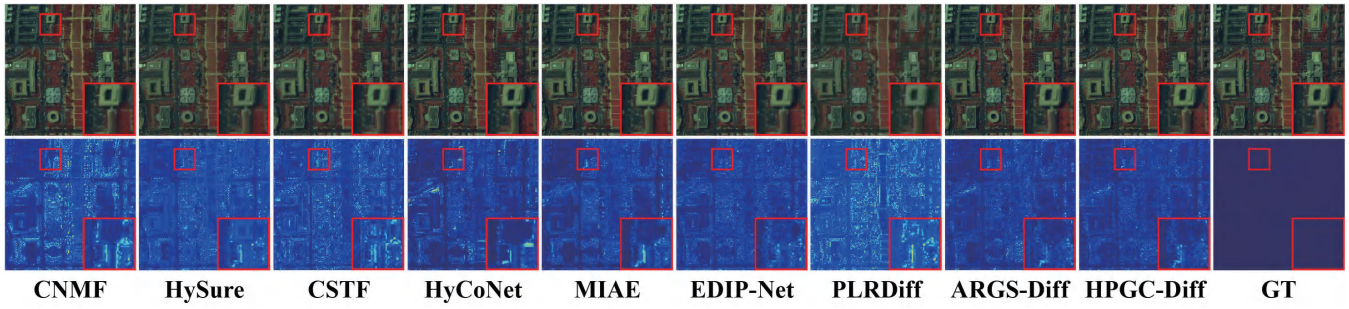


Fig. 7. Qualitative comparison of different fusion methods on the WDC dataset. The top row displays the visual fusion results (R:60, G:36, and B:10). The bottom row shows the corresponding reconstruction error maps, where darker colors indicate smaller errors.

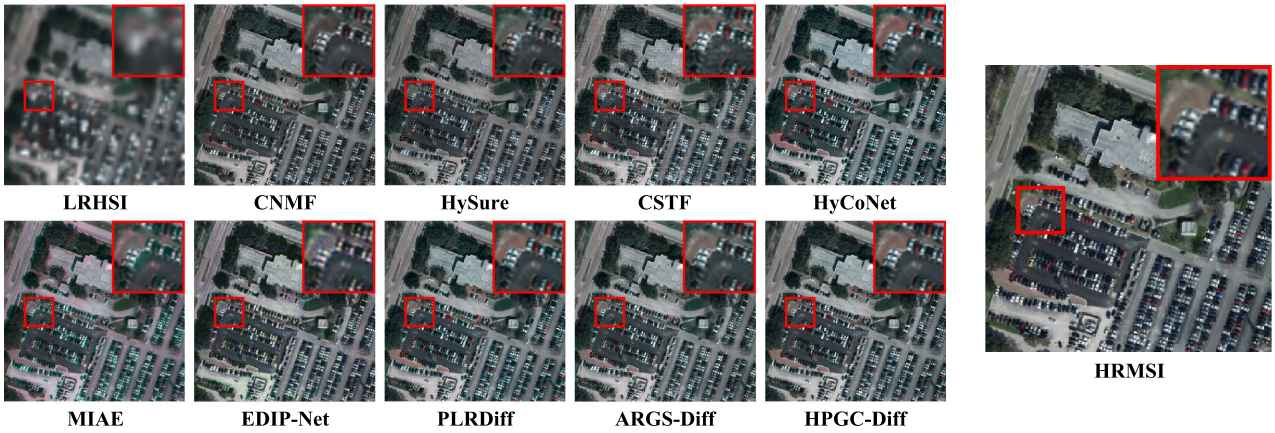


Fig. 8. Qualitative comparison of different fusion methods on a real-world scene from the DFC2018 Houston dataset. For intuitive visualization, the resulting HSIs are displayed as pseudo-color composites using [18, 13, 12] bands.

TABLE III

QUANTITATIVE COMPARISON ON THE WDC. THE BEST RESULTS ARE IN BOLD AND THE SECOND BEST ARE UNDERLINED

Method	PSNR ↑	SAM ↓	ERGAS ↓	SSIM ↑	RMSE ↓
CNMF	32.01	4.85	4.15	0.891	0.0216
HySure	30.81	4.35	4.54	0.905	0.0189
CSTF	33.90	4.94	3.81	0.911	0.0203
HyCoNet	34.44	2.91	3.45	<u>0.952</u>	0.0147
MIAE	35.39	2.98	3.25	0.950	0.0137
EDIP-Net	34.36	3.37	<u>3.22</u>	0.941	0.0148
PLRDiff	25.89	7.22	8.56	0.716	0.0271
ARGS-Diff	<u>36.04</u>	<u>2.88</u>	3.47	0.942	<u>0.0134</u>
HPGC-Diff	**37.87**	**2.75**	**2.46**	**0.963**	**0.0127**

remarkable robustness, excelling in the accurate reconstruction of the complex scene, particularly its road networks and vegetation-covered areas. The quantitative evaluation in Table III rigorously substantiates the superior performance of our method. HPGC-Diff achieves a PSNR of 37.87 dB, surpassing the second-best ARGS-Diff by a clear margin of 1.83 dB. Moreover, its SSIM of 0.963 establishes a clear lead over all competing methods, highlighting its exceptional ability to preserve structural consistency and fine spatial textures. This competitive advantage is further evidenced by the PSNR curve revealed in Fig. 5.

4) Real-World Dataset: To validate the effectiveness and robustness of our proposed method in practical scenarios, we performed qualitative comparison experiments on a challenging scene from the DFC2018 Houston dataset. As shown in Fig. 8, this scene contains intricate details such as densely parked vehicles, posing a significant challenge for fusion algorithms. While traditional methods enhance the overall sharpness relative to the LR-HSI, their capability is fundamentally limited in reconstructing complex details. They tend to produce spatial artifacts and fail to reconstruct fine textures, as evidenced by the blurred separation between adjacent vehicles and the loss of edge clarity along the lane markings. Although DL-based methods are designed to learn powerful data-driven priors, they still struggle to achieve satisfactory spatial fidelity. Moreover, methods such as MIAE and EDIP-Net exhibit pronounced spectral distortions, which manifest as significant color deviations in the pseudo-color images. Diffusion-based methods, by contrast, achieve substantial improvements owing to their strong generative and denoising capability. Among these, our proposed HPGC-Diff delivers visually superior results, faithfully restoring the complex urban structures in the Houston scene. As can be seen, the color of vehicles and roofs in our result remains natural and consistent with the high-resolution reference, strongly demonstrating the robustness and superior performance of our proposed method.

5) Different Scale Factors: Beyond the real-world dataset, we further examine the scalability of the proposed HPGC-Diff under more challenging spatial scale factors. To this end, experiments are conducted on the Pavia University dataset with two larger scale factors, 8× and 16×, to evaluate the model's robustness under increasing degradation

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TABLE IV

QUANTITATIVE COMPARISON ON THE PAVIA UNIVERSITY DATASET UNDER VARYING SPATIAL SCALE FACTORS (8× AND 16×). THE BEST RESULTS FOR EACH SCALE ARE IN BOLD, AND THE SECOND BEST ARE UNDERLINED

Method	Scale Factor = 8×					Scale Factor = 16×				
	PSNR↑	SAM↓	ERGAS↓	SSIM↑	RMSE↓	PSNR↑	SAM↓	ERGAS↓	SSIM↑	RMSE↓
CNMF	35.01	4.85	1.57	0.945	0.0190	32.24	6.12	1.03	0.924	0.0256
HySure	36.70	4.25	1.29	0.959	0.0156	35.93	4.15	0.68	0.957	0.0167
CSTF	38.66	3.52	1.06	0.967	0.0128	38.48	3.83	0.54	0.963	0.0135
HyCoNet	38.79	3.02	0.94	0.966	0.0123	37.98	3.43	0.53	0.964	0.0138
MIAE	40.13	3.29	1.00	0.971	0.0117	38.12	3.88	0.59	0.961	0.0142
EDIP-Net	39.77	2.88	0.89	0.968	0.0110	37.12	3.75	0.57	0.956	0.0146
PLRDiff	39.27	3.47	1.02	0.970	0.0122	38.51	3.68	0.54	0.967	0.0131
ARGS-Diff	40.77	3.17	0.83	0.972	0.0107	39.99	3.34	0.47	0.969	0.0115
HPGC-Diff	**42.28**	**2.82**	**0.79**	**0.980**	**0.0098**	**41.57**	**3.01**	**0.43**	**0.979**	**0.0105**

severity. The quantitative results summarized in Table IV reveal that HPGC-Diff consistently surpasses all competing methods across all metrics and scales. Notably, our method achieves PSNR gains of 1.51 and 1.58 dB over the second-best ARGS-Diff at 8× and 16×, respectively. While a performance decline is inevitable with larger scale factors due to the loss of information, our method still demonstrates outstanding performance and remarkable resilience. These findings affirm the superior scalability and reliability of HPGC-Diff.

D. Ablation Study

1) Effectiveness of the Proposed Method: To comprehensively validate the effectiveness of the proposed 1-D U-Net spectral diffusion model and the dual-source spatial model, we set up three variants for the ablation study.

1) Variant1 replaces the proposed 1-D U-Net with a conventional MLP-based network for spectral modeling, following previous work [51].
2) Variant2 ablates the conditional feature, forcing the model to rely solely on the generic prior from the pretrained generators for spatial reconstruction.
3) Variant3 ablates the pretrained generators, forcing the model to rely solely on the conditional feature from the CFEN for spatial reconstruction.

As summarized in Table V, our full model significantly outperforms all three ablated variants across all metrics. The performance drop in Variant1 validates the superiority of our 1-D U-Net in capturing structured spectral distribution compared to a conventional MLP. Similarly, the degradation in Variant2 and Variant3 confirms the critical and complementary roles of both the conditional feature and generic prior in achieving high-fidelity spatial reconstruction.

2) Number of Pretrained Generators: The number of pretrained generators r for the generic spatial prior is a critical hyperparameter that directly influences reconstruction quality. As summarized in Fig. 9, the reconstruction quality exhibits a nonmonotonic behavior, peaking at $r = 3$ before declining. We attribute this to a tradeoff where a moderate ensemble enriches generative diversity, whereas excessive generators introduce structural inconsistency and optimization ambiguity that outweigh the representational gains. Accordingly, we adopt $r = 3$ as the optimal setting.

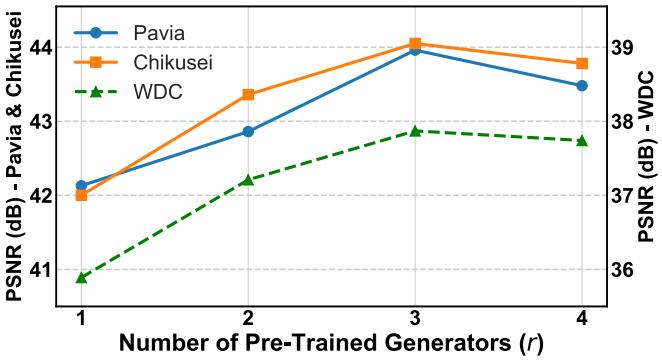


Fig. 9. Analysis of the number of pretrained generators r on the three public datasets.

TABLE V

EFFECTIVENESS OF THE PROPOSED 1-D U-NET SPECTRAL DIFFUSION MODEL AND THE DUAL-SOURCE SPATIAL MODEL

Dataset	Model	PSNR	SAM	ERGAS	SSIM	RMSE
PaviaU	Variant1	43.10	2.50	1.45	0.980	0.0086
	Variant2	43.08	2.33	1.38	0.980	0.0083
	Variant3	40.92	3.29	1.75	0.976	0.0126
	HPGC-Diff	**43.96**	**2.23**	**1.34**	**0.982**	**0.0078**
Chikusei	Variant1	43.19	1.79	2.14	0.974	0.0081
	Variant2	43.24	1.69	1.97	0.979	0.0077
	Variant3	41.21	2.12	2.22	0.976	0.0099
	HPGC-Diff	**44.05**	**1.50**	**1.57**	**0.984**	**0.0069**
WDC	Variant1	36.63	3.51	3.60	0.937	0.0150
	Variant2	36.57	3.28	2.98	0.949	0.0157
	Variant3	34.20	4.06	4.35	0.932	0.0165
	HPGC-Diff	**37.87**	**2.75**	**2.46**	**0.963**	**0.0127**

TABLE VI

ANALYSIS OF THE NUMBER OF SAMPLING STEPS T ON THE PAVIA UNIVERSITY DATASET

Step T	100	200	300	400	500	600
PSNR↑	40.98	42.86	43.37	43.96	43.98	43.37
SAM↓	3.16	2.58	2.30	2.23	2.21	2.42

3) Number of Sampling Steps: The total number of diffusion steps T is a critical hyperparameter that governs the tradeoff between reconstruction quality and inference speed. The results on the Pavia University dataset reported in Table VI indicate that fewer than 400 steps are insufficient to ensure full convergence. Conversely, extending T beyond this threshold

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TABLE VII

ANALYSIS OF GUIDANCE STEP SIZES η_1 AND η_2. ALL VALUES ARE PSNR (DB) REPORTED ON THE PAVIA UNIVERSITY DATASET

η_1 \ η_2	0.01	0.02	0.03	0.04	0.05	0.06
0.02	41.92	41.89	41.77	41.60	41.50	40.94
0.03	43.06	43.18	43.18	42.78	42.70	42.66
0.04	43.28	43.49	43.51	43.02	42.99	42.87
0.05	43.84	43.93	**43.96**	43.31	43.03	42.89
0.06	43.79	43.93	43.83	43.14	43.08	42.81
0.07	43.71	43.83	43.87	43.30	42.88	42.36

TABLE VIII

ANALYSIS OF WEIGHTING PARAMETER γ_2. ALL VALUES REPORTED ON THE PAVIA UNIVERSITY DATASET

γ_2	0.1	0.2	0.5	1	2	5	10
PSNR↑	42.91	43.77	43.86	**43.96**	43.92	43.80	43.33
SAM↓	2.41	2.24	2.24	**2.23**	**2.23**	2.26	2.34

yields diminishing returns, incurring substantial computational overhead with negligible performance gains. Therefore, to strike an optimal balance, we fix $T = 400$ in our final configuration.

4) Sensitivity Analysis of Guidance Step Sizes: The guidance step sizes η_1 and η_2 are key hyperparameters that control the influence of conditional guidance during each reverse diffusion step. To identify optimal settings, we perform a sensitivity analysis on the Pavia University dataset, with results summarized in Table VII. As shown in the table, the peak performance is achieved at $\eta_1 = 0.05$ and $\eta_2 = 0.03$. Notably, the model exhibits strong robustness to these hyperparameters, as a relatively broad range of values around the optimal point yields similarly high performance. Accordingly, we adopt $\eta_1 = 0.05$ and $\eta_2 = 0.03$ as default settings in all experiments.

5) Sensitivity Analysis of Weighting Parameters: The weighting parameters γ_1 and γ_2 control the relative influence of the LR-HSI and HR-MSI fidelity terms within the guidance function. Given that the guidance strength is governed by the ratio between these parameters rather than their absolute magnitudes, we fix $\gamma_1 = 1$ and vary γ_2 to assess sensitivity. The results, summarized in Table VIII, show that peak performance is achieved when $\gamma_2 = 1$. This suggests that an equal weighting of the two fidelity terms yields the optimal balance for reconstruction, and we therefore adopt this setting for all experiments.

VI. CONCLUSION

This article introduces HPGC-Diff, a novel HPGC-Diff framework for unsupervised HSI-MSI fusion. The proposed method effectively integrates the low-rank prior, spectral local correlation prior, and generic prior from pretrained diffusion models to achieve high-fidelity HSIs reconstruction. Specifically, our framework leverages the intrinsic low-rank structure of HSIs, recasting the HR-HSI generation as two separate processes for the spatial structure tensor and the spectral basis. For spectral modeling, we introduce a 1-D U-Net diffusion model, whose hierarchical and convolutional structure provides an inductive bias for capturing the continuous and highly structured nature of spectral signatures. In parallel, the spatial diffusion branch integrates a powerful generic prior derived from pretrained diffusion models with conditional features extracted via a CFEN. To achieve robust reconstruction in an unsupervised setting, we employ an iterative alternating optimization strategy. This approach enables the synergistic evolution of the conditional features and the diffusion-based components under a unified guidance mechanism. Through comprehensive testing across multiple datasets, our proposed HPGC-Diff not only establishes state-of-the-art performance but also demonstrates exceptional robustness.

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